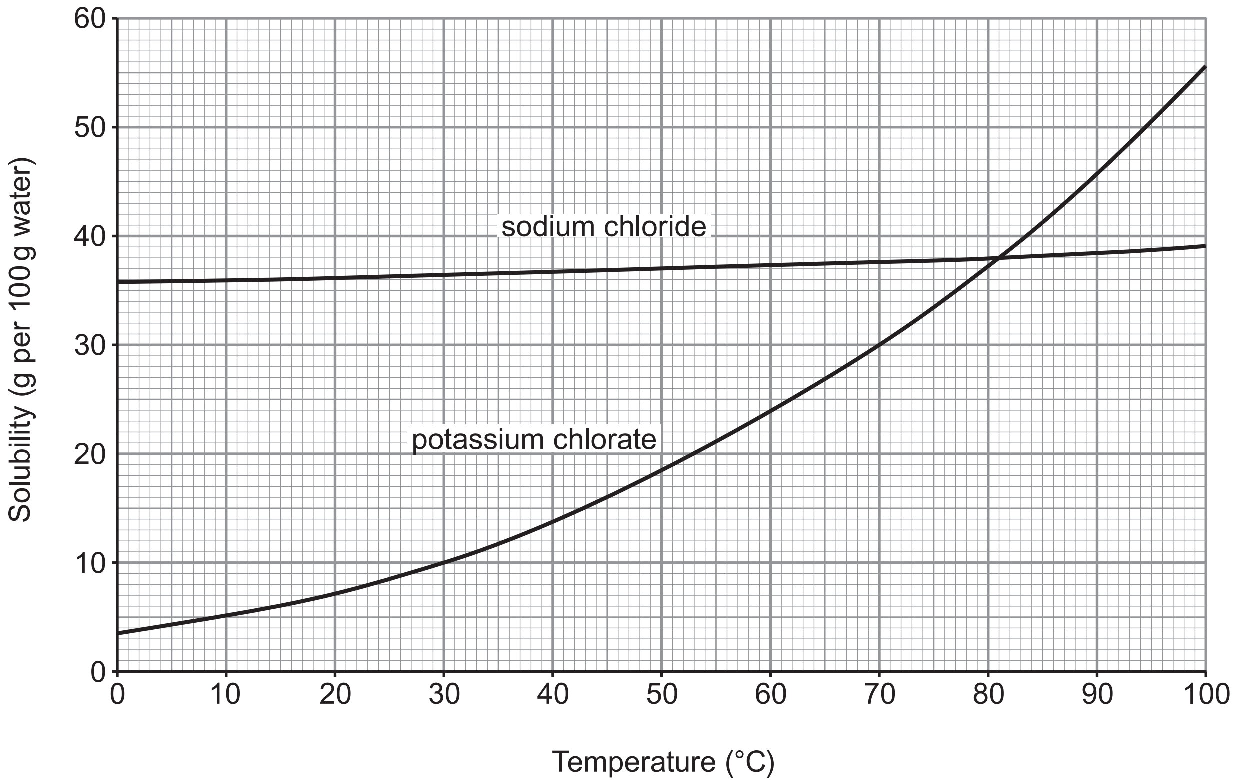
5. *(a)* The graphs below show the solubilities of sodium chloride and potassium chlorate in water at different temperatures.



(i) State the temperature at which the two compounds have the same solubility. [1]

.... °C

(ii) Calculate the mass of solid potassium chlorate that forms when a saturated solution in 100 g of water at 70 °C cools to 30 °C. [2]

mass = ... g

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(b) (i) Calculate the relative formula mass (M_r) of potassium chlorate, $KClO_3$. [2]

relative formula mass = ...

(ii) Calculate the percentage by mass of potassium in potassium chlorate. [2]

percentage by mass = ... %

(c) Potassium chlorate forms potassium chloride and oxygen on heating.

Balance the symbol equation that represents the reaction taking place. [1]

$$2KClO_3 \longrightarrow \boxed{}\,KCl + \boxed{}\,O_2$$

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